EXP:-02 STUDY OF DIFFERE...

16/7/2026 NETWORK CABLES

AIM:

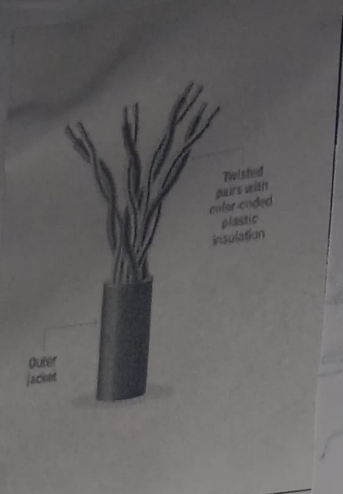
To study and understand different types of network cables, To make your own ethernet cross over cable/straight cable.

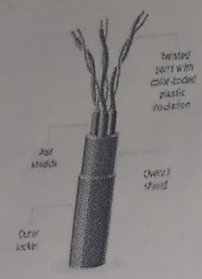
a) Understand different types of network cable.

Different types of cables used in networking are:

1) Unshielded twisted Pairs (UTP) Cables
2) Shielded twisted Pairs (STP) Cables
3) Coaxial Cable
4) Fibre optic

Cable type	Category	Maximum Data Transmission	ADVantage/ Disadvantage	Application
UTP	Category 3 Category 5 Category 5e	10 bps	Adv: *) Cheaps in Cost *) Easy to install as they have a smaller overall diameter. Dis:- More Prone to EMI and noise	10 Base-T Ethernet Fast ethernet Gigabit Ethernet





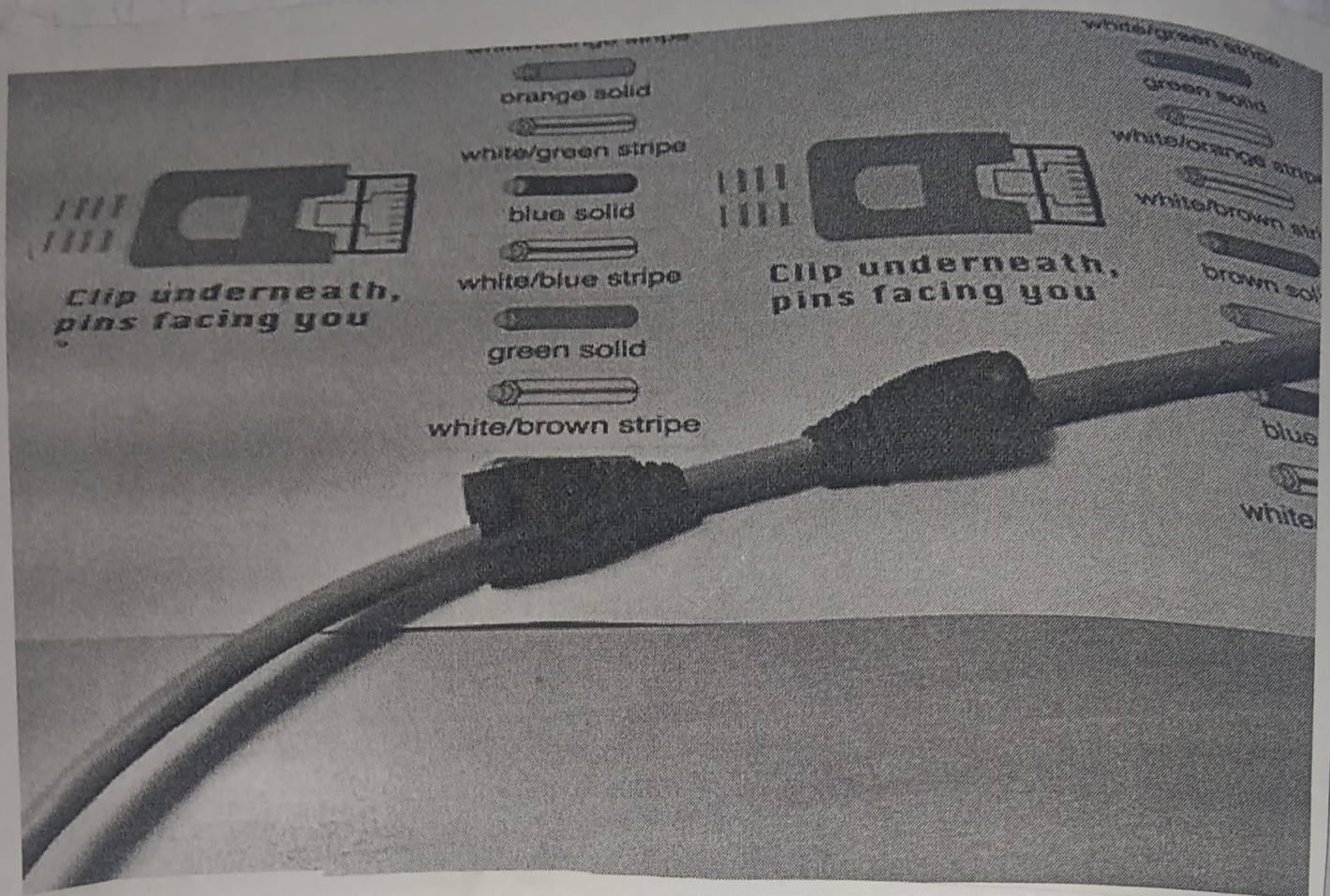
Shielded twisted pair

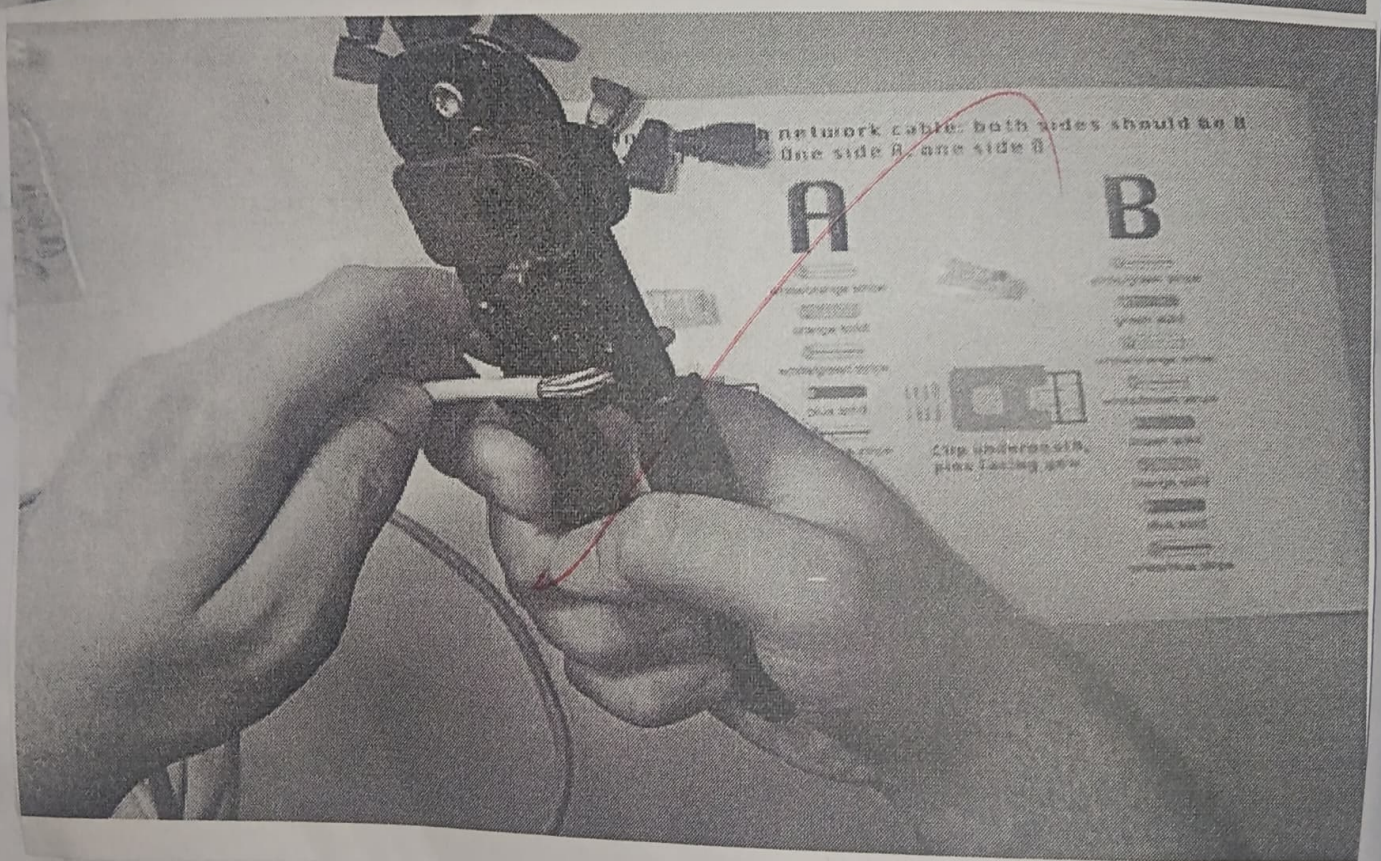
STP	Category 6, 6a	10 Gbps	ADV:- · Shield · Faster than UTP · Less susceptible to noise and methane Disadvantage · Expensive · Greater installation effort	Gigabit 10G ether (55 m) widely used in data centres. gigabet Ethernet 10G. ether (100 m)
SSTP	Category 7	10 Gbps		
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	· high bandwid · immune to interference · Low Loss bandwidth · versatile	· Speed of signal is 500m · Television network · High inter connection
Fibre optics Cable	Single mode multimode	100 Gbps	Adv:- · high speed. · high band width · high security · Long distan	Maximum distance of fibre optics cable is around 100 metres

Make Your Own Ethernet Cross-over cable/ Straight Cable:-

Tools and parts needed :-

· Ethernet cabling. cat 5e is certified for gigabit support, but CAT5 cabling work as well, just over shorter distance.


orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath,
pins facing you
Clip underneath,
pins facing you


A
B

Step 5:- Next, Push the cable right in. the notch at the end of the plug needs to be just over the cables shielding, and if it isn't, that means that you stripped off too much shielding. Simply snip the cables back a little more.

Step6:- After the wires are securely sitting inside the plug, insert the it into the crimping tool and press down. It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step7:- Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable)

to test it, plug it in and attempt to connect two devices directly.

RESULT:-

Thus, different types of network cables were studied successfully, and straight through and Cross over Ethernet cables were prepared and test successfully.